

Chapter 22

Production of Cobalt and Cobalt-Alloy Powders

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INTRODUCTION

Like nickel, cobalt is found in the Earth's crust only in a chemically combined form, save for small deposits found in the alloys of natural meteoric iron. The free element, produced by reductive smelting, is a hard, lustrous, silver-gray metal. Currently, some cobalt is produced specifically from various metallic-lustered ores, for example cobaltite (CoAsS), but the main source of the element is as a by-product of copper and nickel mining. The copper belt in the Democratic Republic of the Congo, the Central African Republic, and Zambia yields most of the cobalt mined worldwide.

Cobalt and cobalt-alloy powders are used in the powder metallurgy (PM) industry predominantly in the production of permanent magnets, superalloys (high-temperature and creep-resistant alloys), and wear-resistant hardfacing alloys. Cobalt powder is also used as the alloying element in tool and die steels, and as a binder in cemented carbides. The cobalt-base alloys are the current materials for application in oral and orthopedic surgery. Cobalt is a significant material for the PM industry due to its capability to improve the high-temperature properties of superalloys, the cutting capability of sintered carbide and high-speed tool steels, and the toughness of high-strength steel.

In this chapter, the following methods of cobalt powder production are discussed: the hydrometallurgical technique, the reduction of oxides, atomization, carbonyl processing, the electrolytic method, the amalgam technique.

HYDROMETALLURGICAL PROCESS

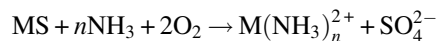
The hydrometallurgical process of cobalt powder incorporates three main processing stages: chemical dissolution (leaching), solution purification, and metal recovery. Because the leaching stage is accomplished in the leach autoclaves, it is also called autoclave processing [1].

Leach Autoclave Method

The leaching autoclave process is based on the ammoniacal pressure leaching of the sulfides to produce soluble nickel and cobalt hexamine sulfate complexes. This process is described in detail in [Chapter 21](#).

The Sherritt (Corefco) refinery in Fort Saskatchewan, Alberta, Canada, was the first successful commercialization of the hydrometallurgical process for the treatment of a nickel sulfide concentrate. This refinery commenced operation in 1954. At present, the primary feed to the Sherritt refinery is nickel and cobalt sulfides, which include 51–56 wt% Ni, 5.0–6.0 wt%

Co, 32–37 wt% S, and 1.0–1.3 wt% Zn. The first stage in the refining process is the leaching of the metal sulfides to soluble amine complexes with the simultaneous oxidation of sulfide sulfur (S^{2-}) to form a series of soluble thiosalts, including sulfate. The general equation for the ammoniacal leaching of sulfides may be represented as:



where M includes nickel, cobalt, iron, copper, and zinc, and n is 2–6.

The typical conditions in the leach autoclaves are represented by the following data: temperature 363–368 K, pressure of 110–120 psig (7.7–8.4 atm), ammonia 80–100 g/L, and ammonium sulfate 150–200 g/L. Before the possibility of reducing the nickel from the leach solution as the pure metal, several solution purification steps are completed. The leach liquor is transferred to the nickel-cobalt separation plant, where cobalt is separated from the nickel as a pure crystalline cobaltic hexamine salt [2]. A flow diagram of this process is shown in Fig. 22.1. The nickel-rich solution remaining after the selective removal of cobalt is directed to the nickel plant for further solution purification processing and final nickel powder production.

Cobalt (III) hexamine sulfate, $[Co(NH_3)_6]_2(SO_4)_3$ is the basis for the separation and purification of cobalt from nickel. The initial step of this separation involves the precipitation of a cobalt-nickel hexamine salt. This precipitation is realized by the addition of ammonium sulfate and ammonia to the leach solution:

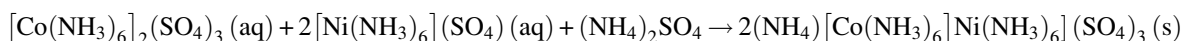
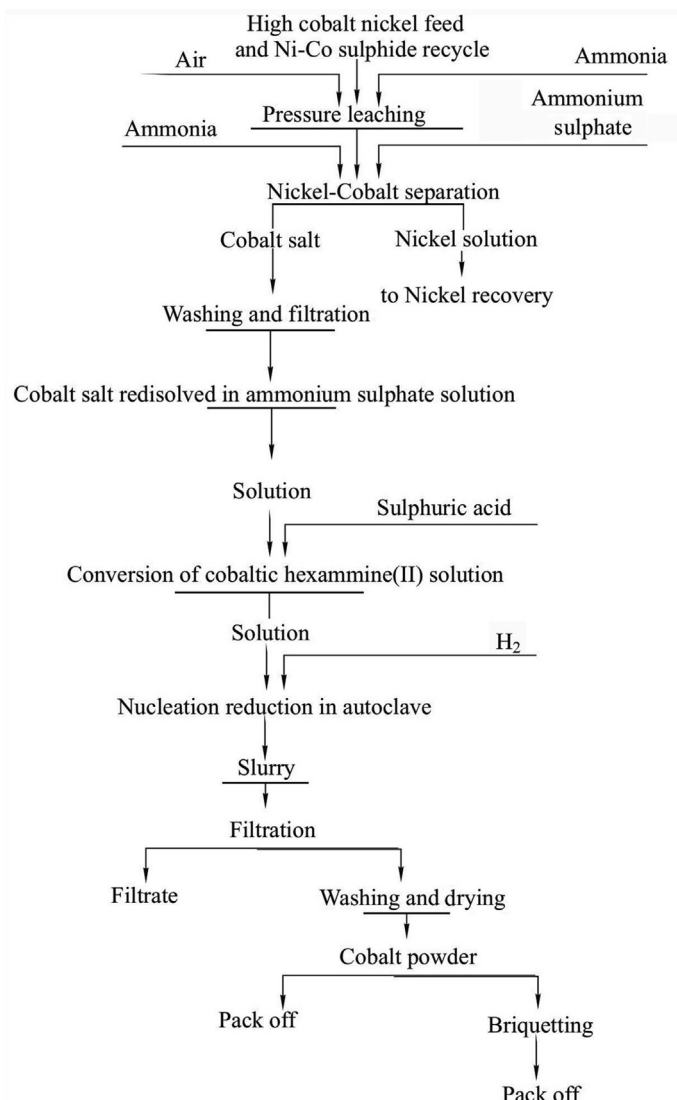
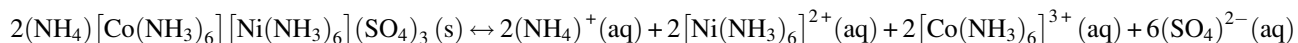


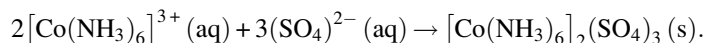
FIG. 22.1 Flow diagram of the Sherritt cobalt refining process.



The relatively insoluble cobaltic hexamine sulfate precipitates in the cobalt-nickel salt while the more soluble nickel hexamine sulfate remains in the solution. This selective nickel removal from the cobalt-nickel hexamine salt results in the isolation of a purified cobalt salt:

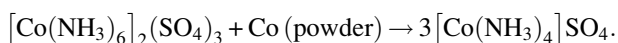


and in the continued reaction



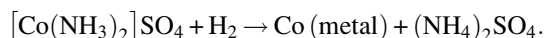
The cobalt hexamine salt is then recrystallized to remove the remaining nickel contamination. After washing and filtration, the pure cobalt salt is dissolved in an ammonium sulfate solution.

A step referred to as “conversion” completes the reduction of the cobalt (III) present in the pure cobaltic hexamine solution to cobalt (II). The conversion of cobaltic hexamine to a cobaltous ammine is accomplished by the controlled addition of metallic cobalt powder, as determined by the following stoichiometry:



The purpose of the reduction process is the chemical reduction of cobaltous ammine to cobalt powder. The reduction of cobalt is a batch operation that occurs in a reduction autoclave under hydrogen atmosphere at elevated pressure and temperature. Cobalt powder is produced in a cycle comprising one nucleation reduction, which refers to the hydrogen reduction process. By this process, fine cobalt seed powder is produced with the aid of a sodium sulfide/sodium cyanide catalyst. After the completion of the nucleation reduction, the solution is decanted from the seed powder, which remains in the autoclave to use as a catalyst. In the course of the cycle, the initially fine seed powder coarsens.

The purpose of the densification processing is to deposit reduced cobalt onto the seed particles until the desired powder density and particle size have been achieved. Every densification is an autoclave hydrogen reduction process, where metallic cobalt precipitates from the solution depositing on the seed powder, by the following reaction:



The rate of the process and its completeness depend on several factors, including the ammonium sulfate concentration, the ratio of ammonia to cobalt, the hydrogen pressure, temperature, and the quantity of seed powder present. At the end of every densification, the powder is deposited, the spent solution is decanted, and the slurry is discharged from the autoclave to a flash tank, where the initial separation of cobalt powder from the end solution is accomplished. After washing the cobalt in a vacuum pan filter to remove ammonium sulfate and drying, the product is packaged as powder or compacted into briquettes and sintered. Typical properties of the Sheritt cobalt powder are shown in Table 22.1.

The kinetics of the reducing autoclave process were studied by Ratner and Dobrokhotov (Gipronickel, USSR, 1966) in the temperature (T) interval 420–520 K, hydrogen pressure $p_{\text{H}_2} = 7\text{--}45$ atm, the ratio of ammonia to cobalt $\text{NH}_3/\text{Co} = 3\text{--}7$, the concentration of cobalt (II), $[\text{Co}^{2+}]_0 = 0.1\text{--}0.8$ mol/L, and at seed consumption 0.11–0.7 mol/L. The kinetic indexes are shown in the first-order autocatalysis equation:

$$[\text{Co}(\text{NH}_3)_n]^{2+} + \text{H}_2 \rightarrow \text{Co} + 2\text{NH}_4^+ + (n-2)\text{NH}_3$$

$$-\frac{d[\text{Co}^{2+}]}{d\tau} = K_0 \frac{[\text{Co}^{2+}]_\tau}{\{[\text{Co}^{2+}]_0 - [\text{Co}^{2+}]_\tau\}} \cdot \frac{p_{\text{H}_2}}{1 + 0.9p_{\text{H}_2}} S \cdot n \cdot 10^{-\frac{2695}{T}}$$

where τ is the time; $S, K_0 = 9.53 \times 10^{-2} \text{ L}^{-1} \cdot \text{mol} \cdot \text{s}^{-1} \cdot \text{cm}^2$; S is the surface of cobalt seed powder, cm^2 ; n is the molar ratio of NH_3 to $\text{Co}(\text{II})$, $[\text{Co}^{2+}]_0$ is the initial concentration of $[\text{Co}^{2+}]$, and $[\text{Co}^{2+}]_\tau$ is the concentration of Co^{2+} at the moment τ .

The reduction mechanism is heterolytic, therefore a seed is very important and there is the possibility of the cladding of the introduced materials by cobalt as well as for making cobalt powders of a desired size, structure, apparent density, fluidity, and specific surface area. The function of the anthraquinone is not so significant; obviously, the ions $\text{Co}(\text{II})$ are not reduced to Co by generating anthrahydroquinone and the catalyst circle of anthraquinone functioning is broken down.

After solution treatment, a period of hydrogen induction is observed, the duration of which decreases with a decrease of the sodium sulfide consumption. Its optimal consumption for fine powder production is 0.01 M. Both the induction period and the treatment duration are decreased in the presence of sodium cyanide. As a result, more dispersed and less agglomerated particles are produced, especially in the presence of surfactant and at a high concentration of ammonium sulfate. The following parameters are operative for the nucleation stage: $T = 423\text{--}473$ K $p_{\text{H}_2} = 1.5\text{--}3.0$ MPa; $(\text{NH}_4)_2\text{SO}_4 > 200$ g/L; and

TABLE 22.1 Properties of the Cobalt Powder by Hydrometallurgical Processing

Property	Unit	Factory, Company			
		Sherritt International Corp.	UMEX	Jedee	
				Extrafine	Ultrafine
Chemical					
Co	wt%	99.9	99.8 ^a	99.9	99.9
Impurities max					
Ni	wt%	0.02	0.15	0.003	
Fe	wt%	0.005	0.001	0.005	0.005
Cu	wt%	0.005	0.0005	0.002	0.002
Ag	wt%	...	0.12	...	...
Al	wt%	...	0.0005	0.003	0.003
Na	wt%	...	0.001	0.005	0.005
Mg	wt%	...	0.0005	0.004	0.004
Mn	wt%	...	0.0005	0.002	0.002
Pb	wt%	...	0.002	0.005	0.005
S	wt%	0.03	0.006	0.005	0.005
C	wt%	0.05	0.18 ^b	0.03	0.03
O ₂	wt%	...	0.7 ^c	...	0.50
Screen size (μm)					
+150	wt%	0–15	...		
–150 +105	wt%	5–25	...		
–105 +75	wt%	5–15	...		
–75 +63	wt%	5–15	...		
–63 +45	wt%	20–45	...		
–45	wt%	10–50	...	100	100
–38	wt%	...	99.98		
Average powder size (FSSS)	μm		0.9	1.01–1.50	0.8–1.00
Apparent density	g/cm ³	2.5–3.5	1.2	...	
Tap density	g/cm ³	...	2.5	...	

^aDetermined by subtraction of nonorganic content from 100% (e.g. Cobalt = 100% – Ni – Cu – Ag – Fe – S).

^bCarbon occurs in the form of residual organic material. 90% of the C is removed during treatment in hydrogen, vacuum, or an inert atmosphere at temperatures up to 673 K.

^cOxygen occurs in the form of residual organic material. During treatment in vacuum or inert atmosphere, up to 75% of the oxygen is removed at temperatures up to 673 K. During treatment in a hydrogen atmosphere, up to 90% of the oxygen is removed at similar temperatures.

NH₃/Co = 2.3–2.5. From other surfactants, the compound containing 1–4 atoms of sulfur at 0.001–0.004 mol of S^{2–} per 1.0 L is of interest.

The main method of powder property variation is to increase the number of densification cycles. The rate of the densification reduction increases with increasing temperature, hydrogen pressure, and seed consumption. Addition of surfactant makes it possible to influence the particle grain size and produce monodisperse powders. However, they are a possible source of powder contamination by carbon and sulfur.

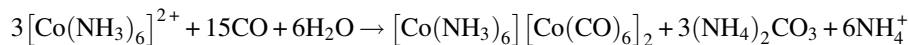
TABLE 22.2 Properties of Composite Powders on the Base of Metal Carbides (Seed Size is 44 μm)

Metal	Part of Coating (%)	Content (%)		Average Size (μm)	Bulk Density (g/cm^3)
		Co	S		
W	81.3	18.6	0.01	14.8	...
Mo	77.9	20.0	0.035	6	1.56
Ti	78.0	19.5	0.09	3.8	1.23
Cr	80.7	18.4	0.046	7.0	2.0

During the period from 1968 to 1982, Outokumpu (Finland) was producing through the leaching autoclave method up to 1500t of cobalt powder per year from secondary raw materials, pyrite concentrate, and semifinished products. The typical conditions in the leach autoclaves were the following: temperature of 450 K and a total pressure of 4.0 MPa. The content of metallic cobalt in the powder produced was 99.8 wt% min.

In 1983, the Impala Platinum Company (Springs, South Africa) built a cobalt powder workshop on the basis of copper-nickel converter matte processing. Double sulfate ammonia salt $\text{MSO}_4(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ was precipitated from the solution after the autoclave extraction of nickel contained 1–3.0 g/L Ni; 0.5–1.0 g/L Co; 400–500 g/L $(\text{NH}_4)_2\text{SO}_4$. The composition of this salt includes 3.7 wt% Co, 11.2 wt% Ni, 0.025 wt% Fe, 0.004 wt% Cu, 0.015 wt% Mn; 60 wt% $(\text{NH}_4)_2\text{SO}_4$. It was dissolved in recycled ammonia solution, purified, and the ratio of ammonia to cobalt adjusted to 2.3–2.5. The solution entering the cobalt precipitation contains 30 g/L Co, 500 g/L $(\text{NH}_4)_2\text{SO}_4$, 20 g/L NH_3 ; max 5 mg/L Ni, 3 mg/L Fe, 1.0 mg/L Cu; 1.0 mg/L Mn. The residual cobalt content in the solution after precipitation is 1–2 g/L.

The use of carbon monoxide as a reducer gas and the treatment of hydroxide suspensions and organic media by hydrogen have been investigated.



Dobrokhotoy (1962) showed the possibility of producing cobalt powder by processing cobalt (II) hydroxide pulp with a content of solid matter 2–3 wt% at 390–420 K, $p_{\text{H}_2} = 2.5\text{--}3.0\text{ MPa}$; in 1 h, powder containing 99.7 wt% Co was produced.

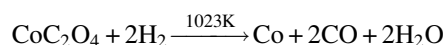
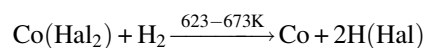
Ultrafine powders (with particles less than 1 μm) are produced by the autoclave method due to the prevention of particle growth. For this purpose, the ammonium sulfate concentration is increased. The initial solution contains a sediment of basic cobalt sulfate as a seed, then the silver sulfate (catalyst) is added as well as an organic surfactant to prevent particle agglomeration. The composition of the powder produced is the following: 0.15 wt% C, 0.75 wt% O_2 , 0.006 wt% S, 0.17 wt% Ni, and 0.15 wt% Ag. Apparent density is 1.1–1.6 g/cm^3 . Specific surface area is 2.0–3.0 m^2/g . Such powder is used in the production of magnetic compounds and as a bond in solid alloy production based on carbides of high melting point metals and diamonds.

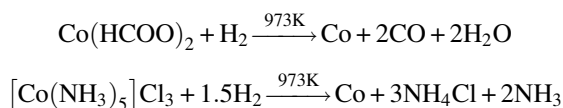
Properties of composite powders based on autoclave cobalt coating of different metal seed types are presented in Table 22.2.

SOLID-PHASE REDUCTION

One of the main applications of cobalt powder is its use as a binder in the production of hardmetals (tungsten carbides). The greater part of this powder is produced by the reduction of cobalt oxide with hydrogen at relatively low temperatures, generally below 1073 K, to obtain the necessary degree of powder fineness. At low reduction temperatures (523–653 K), pyrophorus powder is produced; this property disappears at temperatures above 973 K. To prevent powder from oxidation, it is cooled to 333–343 K in an inert or reducing atmosphere.

The reduction of cobalt salts to metal may be represented as follows:



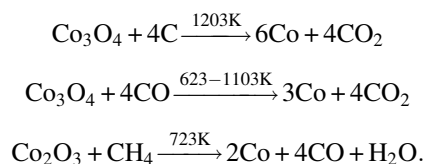


where Hal is a halide anion.

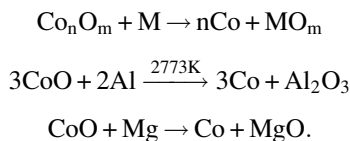
In the work [3], cobalt powder was made through a single-step reduction of cobalt carbonate under a hydrogen atmosphere. SEM microphotographs of this powder (Fig. 22.2) show that the particles in the 0.5–1.0 μm size prevail along with smaller particles up to 100 nm. Particle shape is elongated with a length-to-thickness ratio of about 2 to 1 and their surface looks smoothed (Fig. 22.2A). At a magnification of 50,000× (Fig. 22.2B), it may be seen that the particles that are 150–250 nm in length and 100 nm in thickness are the intergrowing of nanoparticles less than 100 nm in size.

Typical properties of cobalt powders produced by solid-phase reduction are shown in Table 22.3.

Cobalt powder can also be produced by the reduction of cobalt oxide with carbon, carbon monoxide, and methane [1]. However, this technology has not been commercialized. Reduction by these reagents occurs in the following way:



The reduction temperature by carbon is 1203 K, by charcoal and carbon black 923 K, by methane 723 K, and by carbon monoxide from 623 to 1103 K. All processes are carried out in muffle electric furnaces during 1–2 h; powders are contaminated by carbon and contain cobalt carbide. In the case of metallothermic reduction of cobalt oxides, aluminum, boron, magnesium, and silicon are used at temperatures ranging from 2700 to 2900 K:



Cobalt of high purity (99.9% Co) in the form of fine scales or needles is obtained as a result of the treatment of cobalt chloride by aluminum at 903 K in molten NaCl–KCl.

ATOMIZATION

Atomized powders of cobalt-base alloys find a wide use for high-temperature applications. One of the major uses of these powders is hardfacing. This section discusses only hardfacing powders. Gas and water atomization are the most widely used methods for producing hardfacing powders.

The carbide-containing alloys and alloys containing the Laves phase are the two types of commercially effective cobalt-base hardfacing alloys. Typically, a carbide-containing alloy with a nominal composition of Co-28Cr-4W-1.1C is a eutectic alloy containing 16–17 vol% M_7C_3 -type carbides in a cobalt-base matrix alloyed with chromium and tungsten. Laves phase is a kind of topologically close-packed intermetallic compound that has a hexagonal structure similar to

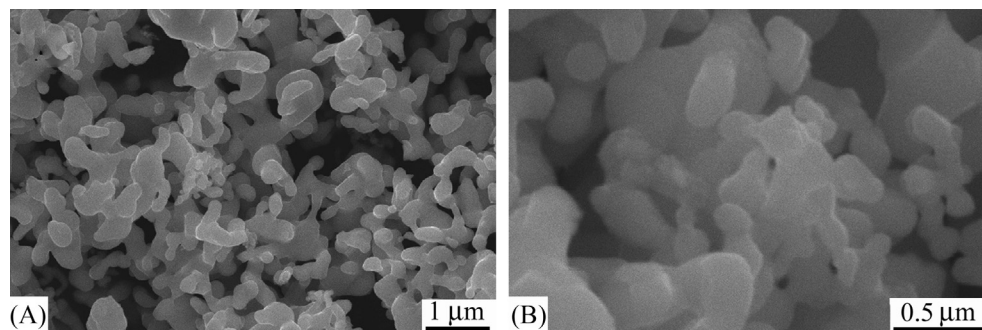


FIG. 22.2 SEM images, at (A) 10,000× and (B) 50,000× magnifications, of cobalt powder made by the solid-phase reduction process. Source: Kurllov AS, Rempel AA, Matrenin VI, Stikhin AS. Morphology and crystal-chemical characteristics of cobalt and nickel nanopowders prepared by thermochemical and electrolytic methods. *Inorg Mater* 2013;49(2):153–8.

TABLE 22.3 Typical Properties of Cobalt Powders Produced by the Solid-Phase Reduction Technique

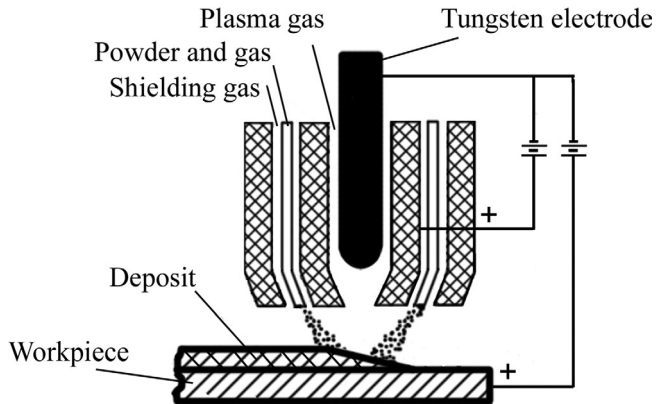
Property	Unit	Value
<i>Chemical</i>		
Cobalt ^a	wt%	99.6
<i>Impurities max</i>		
Nickel	wt%	0.08
Iron	wt%	0.08
Silicon	wt%	0.035
Calcium	wt%	0.020
Manganese	wt%	0.020
Carbon	wt%	0.015
Zinc	wt%	0.010
Sulfur	wt%	0.008
Copper	wt%	0.001
Lead	wt%	0.003
Hydrogen loss	wt%	0.20
<i>Screen analysis (μm)</i>		
+ 150	wt%	0.01
– 150 + 75	wt%	0.04
– 75 + 45	wt%	0.15
– 45 + 38	wt%	0.20
– 38	wt%	99.60
Average particle size	μm	5
Apparent density	g/cm ³	1.8
Tap density	g/cm ³	3.0
^a Hydrogen loss not included.		
Source: Freeman G. Production of cobalt-base powders. In ASM handbook, vol. 7. ASM International Publishers; 1998. p. 179–81.		

M₇C₃ carbides, but the hardness value is less than that of carbides and amounts to 1000–1200 HB [4]. Laves phase materials retain mechanical properties over a wide range of temperatures. Their main uses are for plasma spray hardfacing operations such as plasma spray coating of gas turbine blades. There are two different Laves phase-containing cobalt-base alloys available for hardfacing applications: Co-28Mo-8Cr-2Si and Co-28Mo-17Cr-3Si. These alloys contain at least 50 vol% of Laves phase.

In the microstructure of the carbide-containing alloys, the lightest colored phase is the matrix, which has a face-centered cubic (fcc) crystal structure. The dark phase is M₇C₃; the exact composition is (Cr_{0.85}Co_{0.14}W_{0.01})₇C₃, which has a hexagonal crystal structure. Generally, resistance to abrasive wear is imparted by the presence of carbides whereas the resistance to corrosion and/or elevated temperature hardness retention is imparted by the matrix. The matrix alloy similarly defines galling wear properties.

To minimize oxygen content, gas atomization is the most widely used technique for producing these powders. Some producers prefer vacuum melting and inert-gas atomization (typically nitrogen or argon) to air melting and air atomization. Oxygen content in these atomized powders is significantly below 1000 ppm while the nitrogen level may vary from 600 to 2000 ppm.

FIG. 22.3 Schematic of the plasma-transferred arc hardfacing process.



The plasma-transferred arc hardfacing process is schematically shown in Fig. 22.3. It is a welding process in which the powder is introduced into a combined arc/plasma stream to form a molten pool on the workpiece. The alloy is carried from a powder feeder to the plasma torch in a stream of argon gas, which forms the plasma and is directed from the torch into the arc effluent where it is melted and the fusion is bonded to the base metal. A direct current power source connecting the tungsten electrode and workpiece provides energy for the transferred arc. A second direct current power source connects the tungsten electrode and the arc-constricting orifice supports of the nontransferred arc, which adds the heat of the transferred arc and serves as a pilot arc to initiate the transferred arc.

Powder used for plasma-transferred arc hardfacing process generally [5] ranges from about 150 to 45 μm in size. Coarser powders can be also used because the transferred arc easily melts such powder. Deposits ranging from 0.64 to 3.2 mm thick can be produced rapidly in a single pass.

The composition and hardness of several typical alloys are presented in Table 22.4. The alloys in the Co-Cr-W-C system retain high hardness at elevated temperatures, mostly from carbide and solid-solution strengthening rather than precipitation hardening. As shown, the alloys RCoCrC and RCoCrB are higher hardness modifications of alloy RCoCrA because of higher carbon and tungsten contents. Alloy RCoCrC contains a primary M_7C_3 , a eutectic M_7C_3 plus matrix, and a secondary M_6C . Alloy RCoCrB is similar to alloy RCoCrA, but it has more M_6C and, therefore, contains interdendritic eutectic carbide (chromium and tungsten rich).

Cobalt-base hardfacing powders are unsurpassable for use in combined wear-resistant, corrosion-resistant, and high-temperature applications.

Duflos and Stohr [5] discussed the comparison of the quench rates attained in gas-atomized powders and melt-spun ribbons of Satellite 6 (27.7Cr-4.0W-1.43Si-1.22C-Bal.Co): influence on resulting microstructures. Gas atomization with nitrogen as the atomizing gas used for Satellite 6 was conventional. Cobalt-base alloys were spun to ribbon by melt spinning with a copper roller as the rotating chill substrate. The melt spinning was done in a purpose-built chamber under an inert atmosphere of pure argon or helium. The ribbon width was typically 4 mm and the thickness ranged from 27 to 40 μm depending largely on the fluid flow conditions.

A cellular structure with cell dimension of 0.1–0.2 μm is observed in a 15 μm diameter particle of the gas-atomized powder (Fig. 22.4). The quench rate of such particle sizes, according to authors' estimation [6], is about $10^5 \text{ K}\cdot\text{s}^{-1}$.

The gradation of structure through the ribbon thickness at different levels within the ribbon thickness showed that the major part of the ribbon thickness (20 μm) is represented by a cellular structure (Fig. 22.5). The cell diameter, which varied slightly from the chill zone across the cellular region, was typically of the order of 0.2 μm .

A gas-atomization technology for the production of cobalt-alloy fine powders is used by the Sandvik Osprey Company [7]. The commercial powders, depending on grades (Co 212, Co 502, and Co 90), contain 20–29.0 wt% chromium, 1.0–10 wt% nickel, 2.0–15.0 wt% tungsten, 0.75–3.0 wt% iron, and 0.02–2.0 wt% carbon. The particle shape is predominately spherical. The mean particle size ranges from 6.7 to 12.0 μm .

CARBONYL PROCESSING

At present, the carbonyl method for cobalt powder production is very restricted by low capacity and the unsatisfactory quality of powder [11]. Cobalt carbonyl $\text{Co}_2(\text{CO})_8$ is produced by the action of the carbon monoxide on the fine powder of the freshly reduced cobalt-containing materials at 423–443 K and pressure up to 20 MPa. Reductions of cobalt carbonate

TABLE 22.4 Chemical Composition and Hardness of Typical Cobalt Hardfacing Alloys

		AWS Designation or Trade Name					
Property	Unit	RCoCrA	RCoCrB	RCoCrC	Alloy 21	Alloy 20	Tribaloy T-800
Nominal composition							
C	wt%	1.1	1.4	2.5	0.25	2.5	...
Co	wt%	bal	bal	bal	bal	bal	bal
Cr	wt%	28	29	30	27	32	17
Fe	wt%	3 max	3 max	3 max	2 max	...	...
Mo	wt%	1 max	1 max	1 max	5.5	...	...
Mn	wt%	1 max	1 max	1 max	1 max	...	...
Ni	wt%	3 max	3 max	3 max	2.8	...	...
Si	wt%	1	1.4	1	2 max	17	3
W	wt%	4	8	4	...		...
Nominal microhardness							
DPH	DPH	424	471	577	255	653	653
HRC	HRC	39–42	40–48	52–54	24–27	53–55	54–64
Approximate hardness of microconstituents							
Matrix	DPH	370	420	510	250	540	800 ^a
Particles							
Alloy type		Eutectic	Eutectic	M ₇ C ₃	Eutectic	M ₇ C ₃ M ₆ C	Laves phase
Hard particles	DPH	900 ^b	900 ^b	900 ^b	900	1700	1100
^a Matrix and Laves phase eutectic. ^b Matrix and M ₇ C ₃ eutectic.							
Source: Cabot Corp. AWS—American Welding Society.							

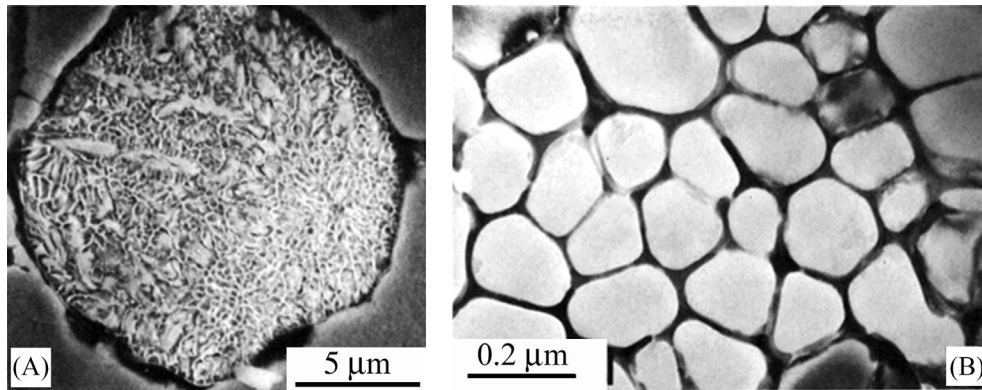
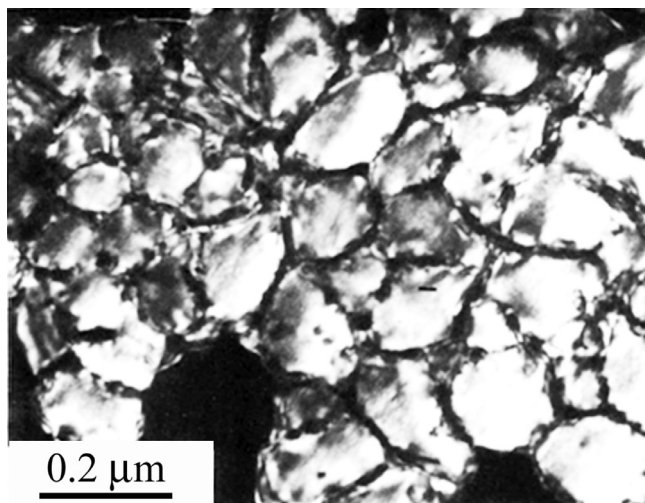


FIG. 22.4 Cellular structure observed in a 15 μm diameter gas atomized powder particle: (A) scanning electron micrograph on a polished section; (B) transmission electron micrograph on a thin foil. From Duflos F, Stohr JF. Comparison of the quench rates attained in gas-atomized powders and melt-spun ribbons of co- and Ni-base superalloys: influence on resulting microstructures. *J Mater Sci* 1982;17(12):3641–52.

FIG. 22.5 Transmission electron microscopy in a Stellite 6 (27.7 Cr-4.0W-1.43Si-1.22C-Bal.Co) ribbon section. From Duflos F, Stohr JF. Comparison of the quench rates attained in gas-atomized powders and melt-spun ribbons of co- and Ni-base superalloys: influence on resulting microstructures. *J Mater Sci* 1982;17(12):3641–52.



in benzene or hexane and synthesis from halide salts in the presence of metallic reducing agents are used. The joint synthesis of nickel and cobalt carbonyls from nickel-cobalt raw material followed by their separation is possible.

Fine black cobalt powder is produced by carbonyl decomposition in a current of carbon monoxide at 423–523 K with subsequent thermal treatment in a hydrogen atmosphere at temperatures ranging from 620 to 1200 K. Passivated fine powder containing approximately 99.99 wt% cobalt and 0.01 wt% carbon is produced by the decomposition of cobalt carbonyl in an inert organic solvent (decalin, silicone and vaseline oil, glycerol, etc.). Organic fluid prevents the contact of the colloid particles formed with air as well as their agglomeration. A protective film that prevents oxidation and powder inflammability is formed due to liquid adsorption on the metallic particle surface.

PRODUCTION OF COBALT AND COBALT-ALLOY POWDERS BY ELECTROLYSIS

Powder is produced from sulfate electrolyte at pH = 5.5–6.0 using soluble cobalt anodes (Fig. 22.6) [1]. Cobalt current efficiency is low (60%) due to low overvoltage of hydrogen on cobalt. A high value of pH solution is necessary to keep cobalt current efficiency above level. Equilibrium material balance in the electrolyzer is regulated by the pH value. The boric acid is introduced as a buffer addition agent. The cathode current efficiency is lower than the anode one so that cobalt concentration in the solution is increased with time. The electrolyte is diluted by condensate to maintain a given concentration of the electrolyte. In the case of solution depletion, the metal dissolution procedure is intensified by the addition of concentrated sulfuric acid.

The anodes are cast from electrolytic cobalt. Sulfate electrolyte is circulated through the electrolyzer in a closed system, including the heat exchanger supporting required temperature in the cells. Electrolysis is kept at current density ranges from 1800 to 2500 A/m² and temperature 333–343 K during about 2–3 h. Powder is precipitated on titanium disc cathodes and then a strippable deposit is gravitated on the tank bottom. The suspended solid is transferred to the settler after the accomplished electrolysis. The clarified electrolyte is returned for electrolysis after filtration while powder is treated by acid solution to wash out hydroxides; it is then washed with distilled water. Powder is dried at 673–683 K during about 3–4 h and reduced at 1173–1233 K during 10 h in a hydrogen atmosphere. After cooling of the vessel with powder in n air environment about 2–3 h and then in s cell with running water (3 h), the powder is milled in ball mills and screened.

The electrolyte is prepared by anode dissolution of cobalt in sulfuric acid solution (at concentrations in the range from 30 to 35 kg/m³) at a current density of 250–300 A/m². Electrolysis is stopped when cobalt concentration has increased up to 18–20 kg/m³, the solution is diluted down to 6–8 kg/m³ of Co, then the pH of the solution is adjusted up to 5.5–6.0 by means of soda and boric acid is introduced as a buffer addition agent (5–10 kg/m³).

Properties of the electrolytic cobalt powder are as follows:

- Chemical composition includes 99.25 wt% Co, max 0.4 wt% Ni, max 0.04 wt% Cu, max 0.2 wt% Fe, max 0.025 wt% S, max 0.02 wt% C, and max 0.1 wt% Cr.
- Granulometric composition consists of the particles max 4 wt% above 71 μm, min 30 wt% (–71 +45) μm, and min 66 wt% below 45 μm.

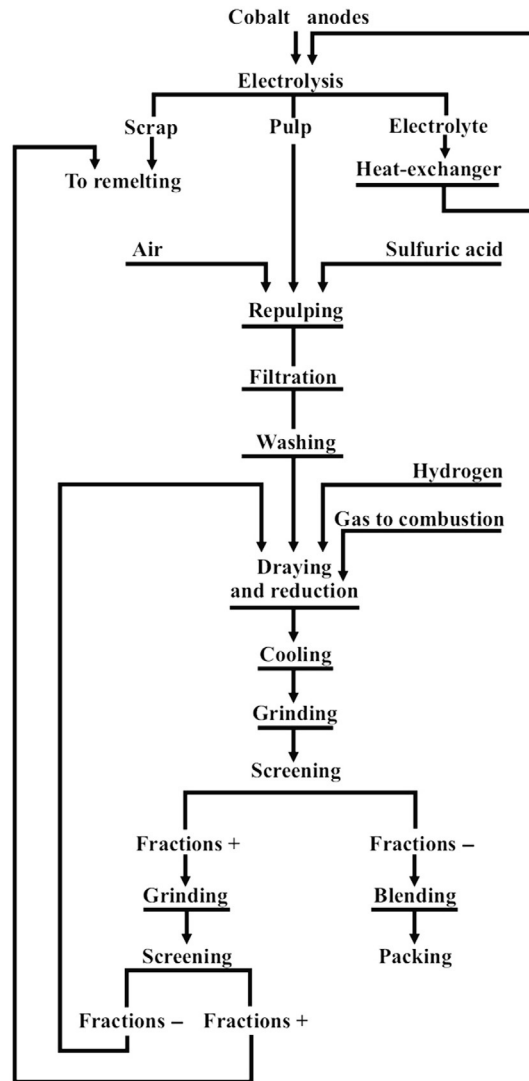


FIG. 22.6 Flowsheet of cobalt powder production by electrolytic processes.

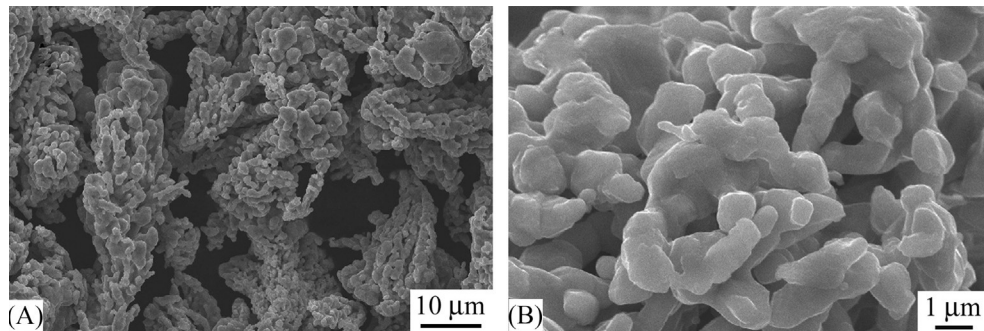


FIG. 22.7 SEM images of the electrolytically prepared cobalt powder.

SEM micrographs of the electrolytic cobalt powder prepared by electrochemical precipitation from an aqueous solution of cobalt and ammonium sulfates [3] showed that its particles were 2–5 μm in length and about 1 μm in thickness and formed a branched dendritic structure (Fig. 22.7A). At a magnification of 10,000×, the large particles are seen to be intergrowths of approximately equiaxed particles ~1 μm in size (Fig. 22.7B).

Comparison of the morphology of the cobalt powders demonstrates the following: at the microstructural level, the solid-phase reduction cobalt powder (Co_{spr}) consists of individual particles (Fig. 22.2A). In contrast, the electrolytic powder has a dendritic microstructure. At a magnification of $10,000\times$, the Co_{spr} (Fig. 22.2A) and electrolytic cobalt (Fig. 22.7A) powders are seen to be similar in particle shape, but the average particle size of the electrolytic powder is about twice that of the Co_{spr} powder. As Curlov et al. [3] discussed, a more in-depth SEM characterization of the electrolytical powder showed that its large particles, about $1\text{ }\mu\text{m}$ in size, were formed by tightly bonded nanoparticles $20\text{--}30\text{ nm}$ in size. Therefore, the electrolytic cobalt powder is nanostructured and has a larger specific surface area than the Co_{spr} powder. These differences in morphology suggest that the electrolytic cobalt powder is more attractive for use in the production of electrochemical power sources.

Various electrolytic techniques of cobalt and cobalt-base alloy powder production are given in Table 22.5.

TABLE 22.5 Production of Cobalt and Cobalt-Alloy Powders by Electrolysis

Method	Electrolyte Composition	Current Density (kA/m^2)	Temperature (K)	Electrolysis Parameters
<i>Cobalt</i>				
WS	CoSO_4 ; $(\text{NH}_4)_2\text{SO}_4$, Na_2SO_4 , H_3BO_3 ;	1.0	300	Particle shapes are needle-like and fan-like dendrites
EDC	5–74 g/L Co, chloride solution, $\text{pH}=1\text{--}6$	0.5–8.0	283–333	Coercive force ranges from 100 to 870 Oe; particle size ranges from 0.5 to $4\text{ }\mu\text{m}$
EDC	23–24 g/L Co; max 25 g/L NaCl chloride solution	3.0–4.0	298	Cobalt current efficiency amounts to 85%–87%
EDC	5–74 g/L Co; chloride solution, $\text{pH}=1\text{--}6$	4.5	298	$\text{Cl}^-/\text{SO}_4^{2-}=1/5$; cobalt current efficiency amounts to 78%–85%; with increasing content of SO_4^{2-} the cobalt current efficiency decreases
SPR ^a	400 g/L NaOH	1.0	353	Raw material is Co_3O_4 or $\text{CoCO}_3 \cdot 2\text{CuO} \cdot 2\text{H}_2\text{O}$; particle size about $0.5\text{--}1\text{ }\mu\text{m}$; electrolysis duration is minimum 4–5 times as large as t_{theory}^b for cobalt oxide, and 7–8 times as large as t_{theory} for CoCO_3
<i>Cobalt-Iron</i>				
AM	62.4 g/L Co; 29.7 g/L Fe; HCl up to $\text{pH}=2$	0.5	313	Anodes consist of 66.3 wt% Co, 33.3 wt% Fe; salts are the sulfates; after heat treatment at 462–473 K for 120–300 min: coercive force range 1700–1850 Oe; particle size is about $0.5\text{--}1.5\text{ }\mu\text{m}$
AM	31.2 g/L Co; 59.4 g/L Fe; HCl up to $\text{pH}=2$	0.5	40	Anodes consist of 33.3 wt% Co, 66.6 wt% Fe; salts are sulfates; heat treatment at 453 K during 60–240 min: coercive force range 1660–1930 Oe; particle size is $0.5\text{--}1.5\text{ }\mu\text{m}$
<i>Cobalt-Tungsten</i>				
WS	21.2 g/L Co; 21.3 g/L Na_2SO_4 ; 5.7 g/L W; 29.0 g/L citric acid; 20.0 g/L $(\text{NH}_4)_2\text{SO}_4$; $\text{pH}=2$	3.0	20–30	Co in the form of sulfate; W in the form of Na_2WO_4 ; cobalt current efficiency amounts to 60%–70%; cathode of Ni; $(\text{Co}/\text{W})_{\text{alloy}} \sim (\text{Co}/\text{W})_{\text{solution}}$ (in molar ratio); alloys up to 20% wt% W are solid solutions; particle shape is globular dendrites with size about $100\text{ }\mu\text{m}$

^aElectrochemical reduction occurs in thin surface film of the dispersoids.

^bCobalt current yield in that system is low, so it is necessary to increase electrolysis duration in compared with t_{theory} – calculated in accordance with Faraday's law for known current.

Production method: WS, aqueous solution electrolysis; EDC, electrolysis in two-layer cell; SPR, solid phase reduction; AM, amalgam metallurgy method (see Chapter 8).

TABLE 22.6 Coercive Force (Oe) at 203 K of the Heat-Treated Cobalt Powder Made by the Amalgam Technique

Temperature of Heat Treatment (K)	Duration of Heat Treatment (min)				Note
	0	20	40	60	
393	530	1080	1110	1110	Without tin addition
408	570	1050	1035	1040	Without tin addition
423	530	1135	1100	1055	Without tin addition
453	565	935	858	845	Without tin addition
303	...	1060	1100	1080	With tin addition
423	...	1116	1065	1065	With tin addition

AMALGAM TECHNIQUES

Electric precipitation of cobalt (or its alloys with iron and nickel) is realized at 313 K from sulfate electrolyte with a cobalt weight-part concentration amount of 94 g/L on agitated mercury cathode in a cell with a soluble cobalt anode (or proper composition alloy) at an interelectrode space that amounts to 1 cm. Anode and cathode current densities are equal at 500 A/m². The required value of pH=2 is achieved by adding hydrochloric acid. Amalgam is heat treated after electrolysis. Thus cobalt particles change their shape from dendritic to needle, or thread-like. Metals (e.g., tin) for counteraction of the particle agglomeration and improving the magnetic properties of suspension are introduced into the amalgam (Table 22.6) [1].

After heat treatment, the amalgam is concentrated by means of filtration and is exposed to orientation in a magnetic field with strength above 50 V/m. This treatment provides mechanically compact forms of hard material. The final mercury removal is accomplished by distillation and the sintered mass is reduced to fine thread-like particles. The particle size ranges from 0.5 to 1.5 μm with the fiber stem diameter around 0.02 μm .

OTHER METHODS

The ultrafine powder is obtained by the reduction of cobalt formate in a vaseline medium (at temperature range 473–513 K during 1 h). This powder is separated by centrifuge, washed by solvent (toluene), and dried. The particle size of the powder ranges from 0.5 to 1.0 μm and the shape is close to spherical by using an ultrasonic field at a frequency about 44 kHz and an intensity of 750 kW/m². This method makes it possible to cover the cobalt particle surface by copper or silver films. For this purpose, the cobalt formate is mixed with glycerol, the silver or copper salts (e.g., nitrates, ammoniates) soluble in the organic phase are added, and the mixture is heated to 423–463 K. Cobalt powder with particle sizes about 0.2–0.5 μm clad with silver or copper is produced.

As powder properties are significantly dependent on the characteristics of the basic reduced solid phase, some decisions for the preparation of the basic raw material before its reduction are proposed [1]. Thus, ammonia salts of cobalt:

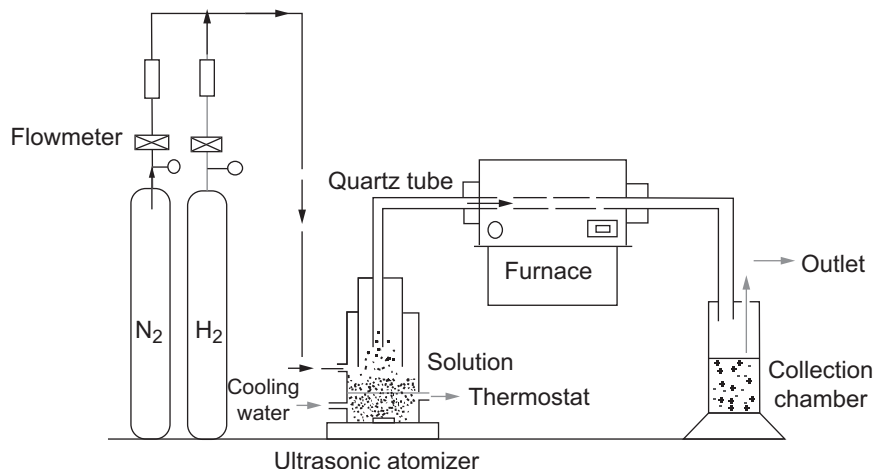
- Are evaporated at 353–373 K to obtain the dry salts.
- Are injected into the chamber at temperature 1173–1473 K to produce cobalt (III) oxide.
- Are treated by a sodium hydroxide solution that results in a precipitate of hydrated cobalt (II) oxide.
- Are treated by carbon dioxide at pressure $p_{\text{CO}_2} = 0.15\text{--}0.5$ MPa, a temperature up to 370 K, and a sediment of alkaline cobalt carbonate is precipitated.

Further, the solid phase obtained is washed, dried, and regrounded. The intermediate product is reduced in a hydrogen current at 670–900 K, and an ultrafine cobalt powder is obtained.

In the case, when cobalt is precipitated from solution in the form of oxalate, the sediment is treated by a sodium hydroxide solution. Prepared cobalt (II) hydroxide is either washed, dried, and reduced in hydrogen, or it is treated by hydrazine at 380 ± 5 K and pH=9.0, or it is calcined and 0.2–2 wt% of tungsten carbide powder is added to the cobalt oxide to prevent powder sintering during reduction by hydrogen at 970–1020 K.

The ultrasonic spray pyrolysis (USP) is a versatile technique for producing various materials in a wide range of composition, size, and morphology [8,9]. Suh and Suslick [10] applied the USP technique to synthesize silica-encapsulated metal nanoparticles. Cobalt nanoparticles were produced by the hydrogen reduction of cobalt chloride in a multistage

FIG. 22.8 Schematic of the laboratory installation for cobalt nanopowder synthesis.



tubular aerosol flow reactor by Jang et al. [11]. Reaction zone temperature, preheating temperature, the mole fraction of CoCl_2 and H_2 , and residence time were considered as key process variables for the control of particle size and size distribution. Ranging from 50 to 78 nm in average diameter, cobalt nanoparticles with narrow size distribution were synthesized throughout the above-mentioned experiments.

Spherical cobalt powders were made by USP of cobalt nitrate aqueous solutions followed by thermal decomposition of generated aerosols in a hydrogen atmosphere [12]. The reaction for the formation of cobalt metal from cobalt nitrate can be described as

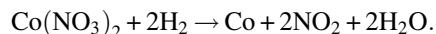
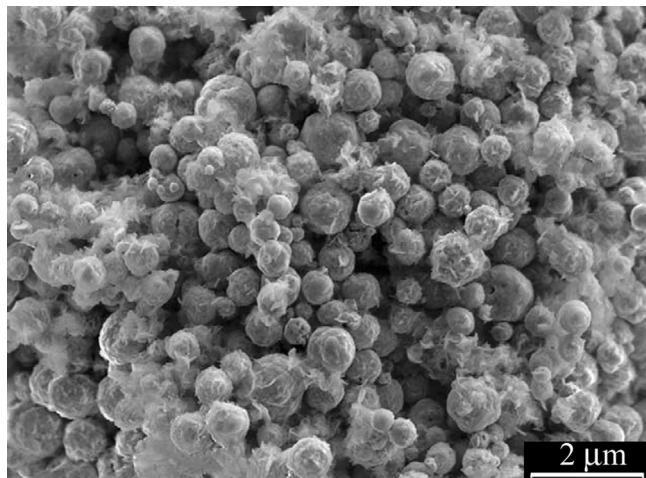


Fig. 22.8 shows the schematic of the laboratory installation for cobalt nanopowder synthesis. A purified leach solution prepared from cemented tungsten carbide scrap using nitric acid was utilized as the starting material [13]. The hydrogen reduction was done at 973–1173 K starting from cobalt nitrate solutions with concentrations of 0.08 mol Co/L and 0.04 mol Co/L. A cobalt nitrate aerosol of about 4 μm was obtained in an ultrasonic atomizer with a frequency of 800 kHz. The liquid feed rate amounted to between 12 and 14 mL per hour. The aerosol was transported by H_2 -carrier/reduction gas to an electrical heated furnace. Under spray pyrolysis conditions in a hydrogen atmosphere and at a flow rate of 1 L/min, the dynamic (continuous) reduction took place in the quartz tube reactor during about 1.0 s.

The recovery of cobalt powder and tungsten carbide from a cemented carbide scrap SEM micrograph of the synthesized cobalt powders at 1073 K by cobalt nitrate solution of 0.08 M Co/L is shown in Fig. 22.9. The cobalt powder particles are almost spherical with an uneven surface; some agglomeration of particles is particularly present.

FIG. 22.9 SEM micrograph of the synthesized cobalt powders at 1073 K by a cobalt nitrate solution of 0.08 M Co/L. From Sebahattin G, Srećko S, Bernd F. Synthesis of nanosized spherical cobalt powder by ultrasonic spray pyrolysis. *Mater Res Bull* 2006;41(10):1882–90.



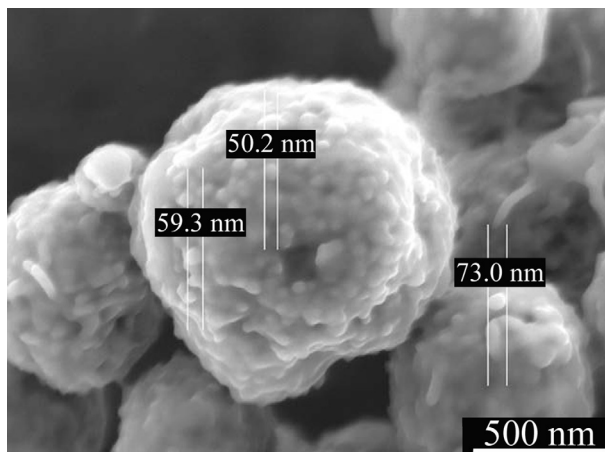


FIG. 22.10 SEM micrograph of ultrasonic spray pyrolysis cobalt powder obtained at 973 K with a precursor cobalt concentration of 0.08 mol/L (magnification 50,000 $\times$). From Sebahattin G, Srećko S, Bernd F. *Synthesis of nanosized spherical cobalt powder by ultrasonic spray pyrolysis. Mater Res Bull* 2006;41(10):1882–90.

The authors [12] discussed that, under the same conditions, the reduced concentration of cobalt in the precursor solution decreases the particle size and uniformity of the powder. The increase in the number of larger particles with cobalt concentration appears to indicate that they are produced by the coagulation, surface reaction, and sintering. However, smaller particles that seemed to be grown by the coagulation and surface reaction are observed at low cobalt concentrations. In difference to nonagglomerated particles by ultrasonic spray pyrolysis nanosized primary particles prepared in the gas phase [9] form aggregates because of interactive forces between particles. Such aggregates change into large particles when the temperature in the gas phase is high enough to sinter the aggregates. Thus, uniform nanoparticles can be produced by spray pyrolysis at proper conditions using uniform drops of precursor salts with low vapor pressure. Fig. 22.10 presents a scaled SEM picture of particles received from 973 K, 0.08 mol/L cobalt nitrate solution by USP. The particles between 50 and 73 nm seen in Fig. 22.10 constitute one big spherical particle of cobalt powder.

Mechanical alloying for the cobalt-alloy powder production of high-entropy alloys (HEAs) is predominantly used [14,15]. In recent times, high-entropy alloys (HEAs) have been the topic of extensive research as these materials are promising candidates for many challenging applications, including tools, molds, and functional coatings. Exceptional properties such as elevated temperature strength, corrosion, and wear resistance result from the unique microstructure [16].

Eißmann et al. [17] reported that the high-entropy alloy CoCrFeMnNi was produced by argon gas atomization. The resulting spherical powder with nearly ideal equimolar composition was subsequently divided. Selected powder fractions were then compacted by two different methods: spark plasma sintering and pressureless sintering. After spark plasma, sintering dense and homogeneous samples are obtained. No change in composition compared to the initial powder could be detected. The sample showed a single-phase microstructure with fcc structure. The density was up to 98% of theoretical value. Pressureless sintering led to samples with a maximum density of 87% for the finest fraction. The achieved hardness of the spark plasma sintering samples was approximately 169 HV and the elastic modulus 198 GPa. The tensile strength ranges from 571 MPa to 613 MPa and the elongation from 27% to 37%, depending on the sintering temperature, while the bending strength was determined to be approximately 920 MPa and the outer fiber strain more than 30%.

HEALTH AND ENVIRONMENTAL PROTECTION MEASURES

Cobalt metal fumes and dust can affect the body if they are inhaled or if they come in contact with eyes or skin. Therefore, they cause irritation of the nose and throat. The current OSHA standard for cobalt metal fumes and dust is 0.1 mg/m³ averaged over an eight-hour work shift. The American Conference of Governmental Industrial Hygienists has recommended a threshold limit value (TLV) for cobalt metal fumes and dust of 0.05 mg/m³. The long-term TLVs of the powders in a workplace atmosphere under the CIS Standard (GOST 12.1.005–88) are (in mg/m³) for: Ni, 0.05; Co, 0.5; Mo metal, 0.5; and W, 6.0.

Metal carbonyls are toxic with volatile ones such as nickel and cobalt carbonyl being particularly dangerous. Nickel carbonyl is a carcinogen as well as being flammable and reactive. Details of nickel carbonyl dangers and measures for health and environmental protection are considered in Chapter 6.

Dicobalt octacarbonyl, $\text{Co}_2(\text{CO})_8$, is especially dangerous in cobalt carbonyl processing. The major health hazards include a potentially fatality if inhaled, allergic reactions, decreased pulmonary function, and dyspnea.

The TLVs of the carbonyls in the workplace atmosphere under the CIS Standard (GOST 12.1.005–88) are (in mg/m^3): $\text{Ni}(\text{CO})_4$, 0.0005 (≈ 0.0004 ppm); $\text{Co}_2(\text{CO})_8$, 0.01; $\text{Mo}(\text{CO})_6$, 1.0; and $\text{W}(\text{CO})_6$, 2.0. The admissible concentration limit of cobalt salts in drinking water is 0.01 mg/L .

Contact of cobalt dust with strong oxidizers may cause fire and explosion. Dicobalt octacarbonyl $\text{Co}_2(\text{CO})_8$ is a non-combustible solid, but by decomposing in air, it forms flammable carbon monoxide that is emitted during decomposition.

Mercury, involved in the amalgam technique, is a toxic chemical. Mercury can affect the body if it is inhaled or if it comes in contact with eyes and skin. It may enter the body through the skin. The current OSHA legal airborne permissible limit is 0.1 mg/m^3 averaged over an eight-hour work shift whereas the National Institute for Occupational Safety and Health (NIOSH) has recommended for cobalt metal fume and dust a TLV of 0.05 mg/m^3 for 10h of work. According to the CIS Standard (GOST 12.1.005–88), the maximum permissible concentration of mercury fumes in the workplace during an eight-hour shift and ultimate nonrecurrent are 0.01 and 0.05 mg/m^3 , respectively.

Implementation of enclosed operations, local exhaust ventilation, and general ventilation must be done to meet the requirements of ensuring the workplace exposure limit. A complex of measures for health and environment protection contains personal protective equipment. Details of safety engineering in the production of powders are considered in Chapter 27.

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